

ANALYTIX PLATFORM

Enterprise Inputs & Outputs Reference Guide

Document Type: Product Technical Overview

Classification: CONFIDENTIAL - Internal & Client Review Only

Target Environment: Microsoft Azure Marketplace Integration

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Section 1: Customer Data Inputs

AnalytiX integrates directly into existing life-sciences IT landscapes. By leveraging advanced data connector adapters and schemas, the platform acts as a unified ingestion fabric. It seamlessly handles structural chemistry, high-throughput biology, enterprise databases, and unstructured scientific files.

Category	Supported Sources & Protocols	Ingested Data Types
Laboratory Systems	• ELN: Benchling, Dotmatics, Signals Notebook • LIMS: LabWare, LabVantage, STARLIMS	Experimental protocols, sample runs, inventory mappings, user audit logs.
Enterprise Databases	• Relational & Warehouses: PostgreSQL, Oracle, SQL Server, Snowflake, MySQL	Federated query views, historical screening data, assay metadata databases.
Scientific Files	• File Ingest: CSV, Excel, JSON, XML, FASTA	Raw assay values, plate layouts, genomic/nucleotide data, physical descriptions.
Enterprise Integration	• Connectivity: REST APIs, SOAP APIs, FTP/SFTP, Cloud Storage (AWS S3, Azure Blob, GCS)	Robot file drops, vendor data deliveries, automated system-to-system syncs.
Scientific Data	• Biological & Chemical Entities: SMILES, SDF, Sequence files (DNA, RNA, Protein), Assay outputs	Chemical structures, sequence alignments, inhibition values, metadata tags, audit ledgers.

Section 2: Platform Data Processing

Upon data ingestion, the AnalytiX platform executes sequential pipelines to normalize, register, link, and analyze scientific entities in a regulated, high-performance environment.

- **Data Ingestion & Validation:** Performs type verification, validation rules checks, and structural normalization on compounds and sequences.
- **Schema & Metadata Discovery:** Automatically inspects connected databases to map structures and index attributes into a dynamic EAV Metadata Catalog.
- **Federated Query Execution:** Translates queries across distributed databases (e.g. Snowflake + local PostgreSQL) without data replication.
- **Scientific Analysis:** Runs RDKit chemical descriptors calculations, global/local sequence alignments, and dose-response curve regressions (4PL models).
- **Workflow Automation & AI:** Executes multi-stage pipelines (e.g., query candidate -> filter ADMET -> dock molecule -> request compliance signature) with AI Copilot support.
- **Lineage & Auditing:** Maintains complete end-to-end data lineage from source data connector to final fit, writing to a cryptographically linked SHA-256 ledger.

Platform Processing Architecture Flow



Section 3: Platform Data Outputs

AnalytiX outputs are tailored to support scientific discoveries, project leadership review, and strict regulatory inspections.

Scientific Analytics: Dose-response sigmoidal curves (IC50/EC50 fits), molecular similarity matrices (Tanimoto coefficients), genomic sequence identity grids, R-group distributions, and correlation maps.

AI Copilot Outputs: Natural language answers to databases queries, automated study summaries, compound optimization recommendations, and workflow template generations.

Enterprise Reports: High-fidelity PDF compliance files, structured Excel summaries (with raw assay metrics), PPTX presentations for project review, and regulatory audit-ready logs.

Compliance Records: FDA 21 CFR Part 11 electronic signature validations, SHA-256 cryptographic audit logs, and interactive interactive visual data lineage tracking reports.

Operational Alerts: Workflow run status notices, automated approval request notifications, and scheduled pipeline reports delivered via Webhooks, Email, or Slack.

Section 4: Enterprise Business Benefits

The platform maximizes value across all departments, establishing a modern, compliant, and data-driven discovery pipeline.

RESEARCH SCIENTISTS <i>Accelerated In Silico Discovery</i> • Saves hours of database manual querying and scripting. • Correlates chemical structures with biology assays automatically. • Natural language interface allows complex queries without SQL.	RESEARCH MANAGEMENT <i>Data-Driven Portfolio Direction</i> • Real-time project dashboard trackers and KPIs. • Fast candidate optimization cycle analysis. • Centralized decision support for candidate selection.
COMPLIANCE TEAMS <i>Audit-Ready Transparency</i> • Strict FDA 21 CFR Part 11 compliant digital logs. • Complete lineage mappings from raw source to final charts. • Immutable cryptographic audit trailing blocks.	IT & INFRASTRUCTURE <i>Secure, Integrated Ecosystem</i> • Connects directly to Snowflake, PostgreSQL, ELN, and LIMS. • Granular RBAC, SSO, and OAuth standard integrations. • High availability containerized microservices structure.

Section 5: Sample Client Journey

Scenario: A biopharma client connects their PostgreSQL screening database, Benchling ELN protocols, cell-line CSV results, and a set of FASTA sequences.

AnalytiX Automated Sequence:

1. **Connects:** Securely establishes read-only data connections in minutes.
2. **Discovers:** Auto-maps database tables and extracts custom metadata fields.
3. **Relates:** Link assay data plates directly to compound chemical profiles using SMILES mappings.
4. **Analyzes:** Computes dose-response parameters (IC50 curves) and registers chemical structural descriptors.
5. **AI Grounds:** Scientist Copilot translates natural language questions into secure queries, returning analog candidate structures and compliance PDF summaries ready for review.

Section 6: Visual Infographic Summary

The AnalytiX value chain transforms customer data investments into validated business and scientific outcomes.



ELN & LIMS Systems
Databases & Files
REST/SOAP APIs
Raw Structures

Schema Discovery
Data Standardization
Federated Engine
Data Lineage Map

AI Scientist Copilot
4PL Dose-Response
Sequence Alignments
Compound Analytics

Faster Lead ID
100% Compliance
Data Modernization
Low IT Silos